EXPERT TAKE FOLLOW THE MONEY

What's the Renminbi Got to Do With it?

A closer look at the impact of China's currency on China's export growth.



Sean Yong/Reuters

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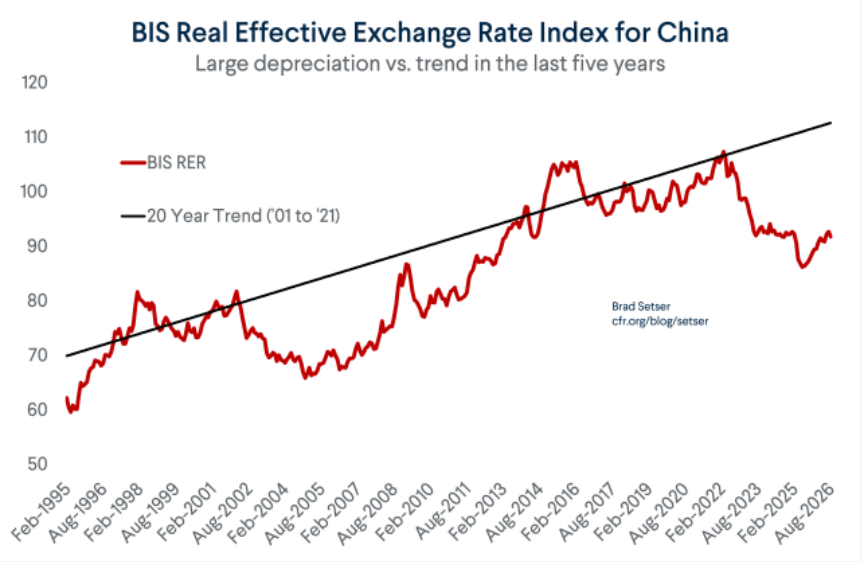
There is an argument—made in the recent Geneva Report and by various former IMF officials—that China's current export surge is the simple consequence of the housing downturn and the resulting weakness in Chinese internal demand.*

The weakness in the RMB in this story is entirely incidental to the increase in exports, which is, more or less, a mechanical response to weak internal demand.

Cannot sell excavators at home? Export them.

Built more modern car capacity than your internal market supports? Export that too…

The fall in the nominal RMB over the course of 2022 and the first half of 2023 in nominal and read terms was simply a function of domestic macroeconomic weakness, and a growing rate differential. It wasn't an active policy choice by China's central bank.



Now, some might note that China's currency doesn't always move with rate differentials. The RMB didn't consistently get pulled up when Chinese rates were higher than U.S. rates (2002-2020 generally speaking), and while it did fall when rate differentials moved against the RMB in 2022 and 2023, it then stabilized and has even appreciated modestly since the middle of last year. That appreciation hasn't been big enough to correct the deep undervaluation of the RMB, but it came at a time when the PBOC policy rate (the reverse repo rate) was well below the U.S. policy rate.

China doesn't have an open financial account. As a result, equilibrium in the foreign exchange market is thus driven far more by exporter conversion and current account flows than by rate differentials and financial flows.** That, together with China's ongoing management of the exchange rate and a trade surplus that can generate signi[illegible]nflows, means that the

value of the RMB is much more of a policy variable under the independent control of the Chinese authorities than a simple byproduct of monetary and fiscal policies.

The modern IMF is reluctant to accept this conclusion. Its surveillance has generally ignored China's currency management (even if that management is relatively obvious; see the data on FX settlement) and its recent comprehensive assessment of China's economy completely ignored the balance of payments and exchange rate management.

But it is pretty obvious to the market, even if the PBOC, rather absurdly, insists that "market forces play a decisive role in determining the RMB exchange rate."

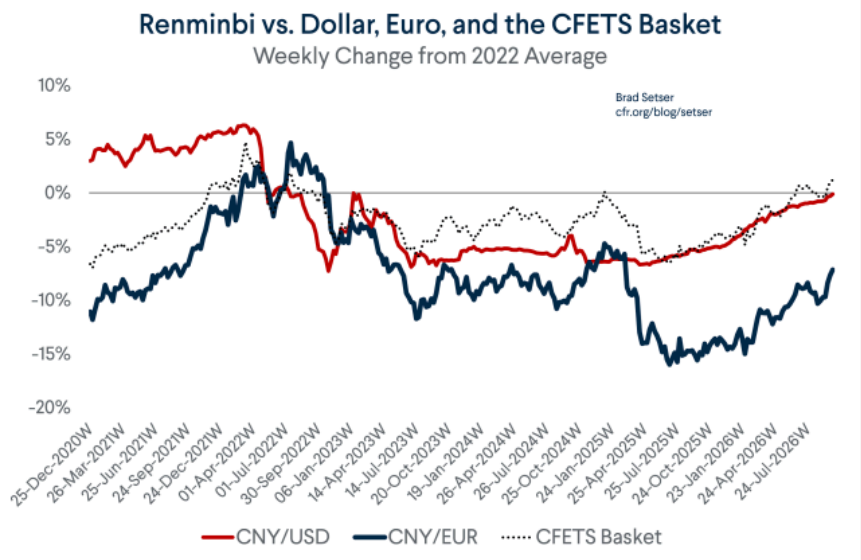
I doubt the PBOC can win this argument. I agree with Mark Sobel:

"The renminbi is still heavily managed and the authorities have acted to slow appreciation. The paper states that the PBoC phased out 'regular' foreign exchange intervention in 2017. But it has many means at its disposal to achieve the same effect through extensive capital controls, administrative guidance, signalling at the fix and deploying state-owned commercial banks. These means are heavily used. Thus, the word 'regular' must be taken with a pinch of salt. The com[illegible] to Japan, which

intervenes occasionally, is spurious – the yen largely floats freely in integrated global markets. The same cannot be said for the renminbi."

In reality, the PBOC's daily fix now plays the decisive role in the currency's path, and the market is often a bet on how the PBOC's management of the yuan will evolve.

That is clear from the lack of volatility in the yuan's rate against the dollar. The yuan's move down against the dollar in 2022 was a controlled 10 percent move, and the appreciation over the last 16 months has been even more controlled crawl.



And the market is clearly massaged with substantial help from the state banks. In fact that the pattern of changes in the monthly FX settlement data looks exactly like the pattern of intervention of a central bank managing its cur round a tight band (the

state banks in practice buy FX to keep the CNY from testing the strong side of the band, so there is an asymmetry in how the CNY is managed)



None of this would matter much if China's economy was small. But that obviously isn't the case. Nor would this matter much if China's exports didn't respond to changes in the RMB's real (and nominal) value. But that is also not the case.

There is abundant evidence that China's exports do respond to moves in the currency. Periods of RMB weakness have consistently been associated with periods of especially rapid Chinese export growth.



Chancellor Merz and President Macron got this right, even if macroeconomic issues haven't yet figured prominently in the EU's dialogue with China.*** As they note in their open letter to the European Commission:

> "An appreciation of undervalued currencies towards a value more in line with fundamentals would help support the rebalancing of trade accounts with the EU in the short and medium term … European policy discussions focused on forging a common understanding of these exchange rate issues and their interaction with global imbalances would be useful."

The best evidence that exports responded to changes in the currency, and not simply to changes in the path of China's internal demand, is the acceleration in China's export growth after the RMB's depreciation over the course of 2022.

This is the killer chart.



Now, you might ask: what of the acceleration in export growth in 2020 and 2021? That was pretty clearly tied to the global increase in goods demand that came with the pandemic and work from home requirements. It was expected to fade when the world economy normalized. But it didn't. By 2023 Chinese export growth again topped global trade growth.

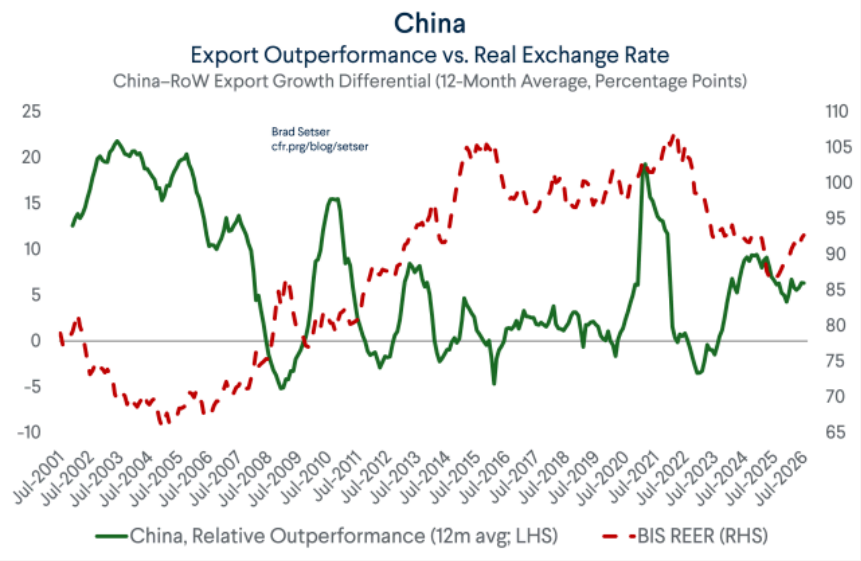


The general expectatio[redacted]the rise in China's goods surplus during Covid would prove to be "transitory

heavily influenced the IMF's work at the time: see the IMF's external sector reports of 2023 and 2024.

I would argue that neglecting the exchange rate was the key reason the IMF's kept forecasting, incorrectly, that China's current account surplus would fall (See this blog with Shahin Vallee). The IMF certainly didn't warn that China's property downturn and the RMB's real depreciation risked a second China shock in its 2022, 2023, or 2024 surveillance reports.

Concretely, the weaker RMB appears to have contributed to a very large outperformance of China's exports relative to global trade. See the following chart



That incidentally also happened after the RMB followed the dollar down in 2003.

In both periods, there [illegible] significant structural

of Chinese industrial policies in the "green" industrial sectors and autos in 2022. In both cses exports were turbocharged when structural shifts that should have resulted in a stronger currency were instead amplified by currency weakness.

No matter. Paul Krugman has noted that a country that is converging technologically with its trade partners should see a trend appreciation in its currency. China got the technological convergence, but not the real appreciation—at least not until this year. And the move since the middle of 2025 has corrected the depreciation of the RMB that occurred when the RMB followed the dollar down on liberation day, but not the structural overvaluation.

Now, there is an argument that exchange rate adjustment is too hot an issue to make it a priority.

The PBOC isn't interested it seems.

And some believe that in the interest of comity and good relations with China, the IMF should mirror the PBOC's views. After all a stronger RMB would slow China's exports and thus slow China's economy at a time when China needs export growth to escape deflation and generate employment. Or so the argument goes. Concerns about "external balance" and the negative global spillovers from China's rising goods surplus should be ignored and the Fund's sole focus should be on poli[illegible]at lead to internal

balance. That provides one rationale for the Fund's ongoing decision to essentially ignore the contentious issue of the PBOC's ongoing exchange rate management.

That argument, however, assumes that China's economy is small and thus China's external surplus can safely be ignored because it isn't a drag on the growth of the IMF's other members.

That clearly isn't true.

And it isn't clear that China is more interested in having a negotiation over its level of basic pension benefits or the size of its fiscal deficit (where the IMF continues to make timid recommendations that would not move the needle—and certainly wouldn't meaningfully reduce China's external surplus) than a discussion over its currency practices.****

Back in the world of fixed but adjustable exchange rates, there was an expectation (not always met to be sure) that surplus countries would allow their currencies to strengthen to bring their economies back into external balance, and that other policy tools would be used to ensure internal balance.

I would argue that China's exchange rate (against both the dollar and the basket) has many similarities to the old world of fixed but adjustable currencies. True, it moves on a co[illegible] path when it moves

between different levels rather than jumping up and down. But the PBOC does frequently target a level, and that targeted level does change over time.

I would further argue that steps like the recent tariff rate quota that will limit China's exports of hybrid cars to Europe to around 350,000 units next year (up from 2025, but well below the 800,000 units of 2026) aren't going to address the underlying imbalance on their own. They can provide a bit of a buffer for specific sectors.

But that buffer, while necessary, won't address the broad-based erosion of manufacturing capacity outside China. This erosion will persist as long as an undervalued RMB sends the entire world a signal to use China as its production base.

The new system of managed trade that is emerging in autos would need to be progressively expanded to an ever-increasing set of sectors if protection is used as a substitute for both domestic reform in China and exchange rate adjustment.

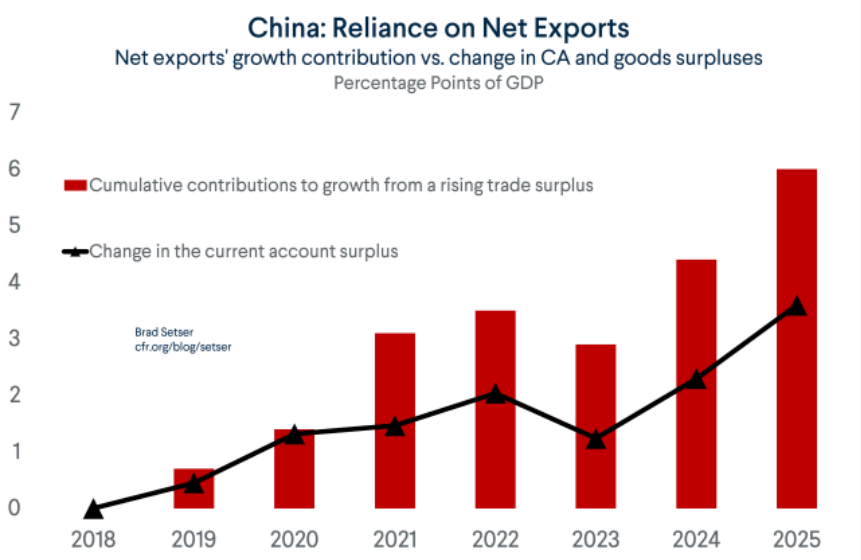
That is now the direction of travel, but it doesn't address the underlying systemic issues.

Incidentally, those who believe the path of internal demand is critical to rebalancing and that exchange rate moves are secondary need to explain how an acceleration in domesti[illegible]and would, on its own,

lead China's external accounts to adjust. One important, stylized fact about China is that domestic demand growth is no longer feeding over to import growth (ex-gold, that is).

Bottom line: China, which needs a 1.5 percentage point contribution from net exports *every year*, is on a collision course with its trading partners without a currency correction that provides the basis for more balanced trade even in the absence of trade restrictions.

And there isn't a realistic path towards balance that doesn't involve a stronger RMB; there aren't, as Paul Krugman once put it, immaculate transfers.



*That downturn shows up in domestic deflationary pressure, and weak job growth/high rates of (at least youth) unemployment [illegible], in other words, isn't in internal balance. I really liked the report from

Kamakshya Trivedi and Hui Shan of Goldman Sachs arguing China needs two tools to hit two targets (internal and external balance).

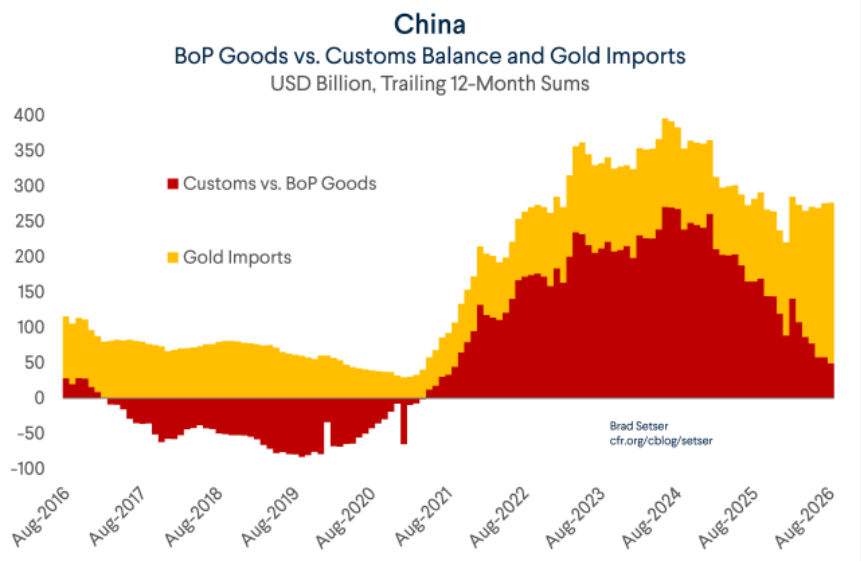
** To be sure, rate differentials matter for exporter conversion—but so does the expected movement of the RMB. The market understands this better than the Fund. The Chen and Suwanapruti (Goldman Sachs) note uses a framework similar to mine to describe the CNY's moves in Q3:

> "In our view, sizeable FX inflows have continued to support RMB appreciation despite unfavorable interest rate differentials, as strong export receipts and increased exporter FX conversions have outweighed portfolio outflows. Our estimated FX conversion ratio rose to 63% in August, compared with around 45% in normal years, likely reflecting a shift in market expectations from RMB depreciation to appreciation."

*** Šefčovič has jurisdiction over trade policy (narrowly defined), but not broader macroeconomic policies. Europe's institutional architecture for exchange rate coordination is not well-developed.

**** The current account surplus is no longer masked to the same extent as in 2023 and 2024, as SAFE has fiddled with its BoP methodology once again (per the Federal Reserve) and the BoP goods surplus is no longer significantly below the customs goods surplus. Of course, there are still problems in the data—the deficit in the investme[illegible]me account most

obviously, but also the recent surge in "customs" gold imports.

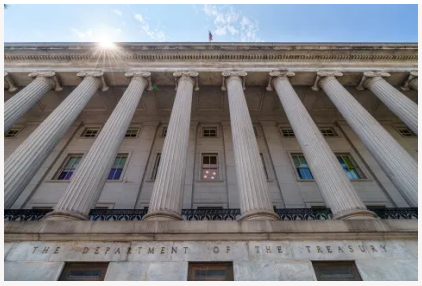


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CHINA ECONOMICS TRADE



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De-reservification, Not De-dollarization

The world isn't moving away from the dollar. It is shifting its dollars from traditional reserves to state banks, pension funds, and other quasi-sovereign investors.



(Bernardo Montoya/Courtesy Reuters)

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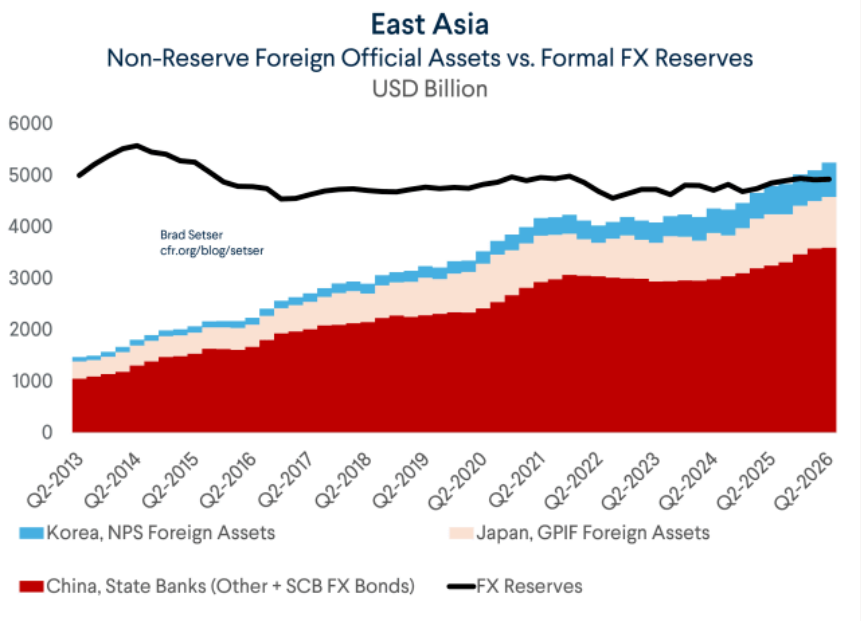
This is a joint blog with Alex Etra, Senior Macro Strategist and head of Flow of Funds at Vanda Research.

The IMF's data on the currency composition of foreign exchange (FX) reserves is scrutinized carefully for any signs that the world is shifting way from the dollar.

Yet it doesn't really matter that much if the dollar's share of reserves has shifted from 56.5 percent to 57 percent. The ink-to-impact ratio of the quarterly reporting on the latest COFER data is all off.

Neither the stock of global reserves nor the stock of dollar reserves has changed much in the last ten years.

The action is elsewhere.



China's state banks (per the Bank of International Settlements) have almost as many foreign assets as the central bank (PBOC).

Japan's Government Pension Investment Fund (GPIF) has almost as many foreign assets ($986 billion) as the government has FX reserves ($1.1 trillion at the end of August). Japan's FX reserves are on the books of the Ministry of Finance and the GPIF is overseen by the Ministry of Health, Welfare and Labor—so these are almost all assets of the Government of Japan or the broader public sector, not its central bank.

Korea's National Pension Service has more foreign assets than the Bank of Korea has FX reserves.

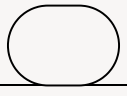
A different dynamic is in play in Taiwan, as the government doesn't ru retirement fund. But in

a world where the hedge ratio of the life insurers has been (and still is) used as a policy tool, the shift in Taiwan's foreign asset accumulation away from the Central Bank of China (CBC) and over to institutions whose activities in the foreign currency market it directly influences broadly fits into the same theme. So too does the PBOC's use of the "macroprudential adjustment factor" effectively adjust banks' net foreign assets in much the same way that CBC regulations adjust lifer hedge ratios.

In the last year, the lifers' hedge ratios—and their direct hedges with the central bank—have been one of the key tools the CBC has used to manage the Taiwan dollar (and guard the lifers' solvency).

Asia's big surplus economies are in a sense just catching up with the world's oil exporters. Norway put its oil surplus into a sovereign wealth fund from the start. Kuwait, Qatar, and Abu Dhabi (the most oil-rich emirate) all also have large sovereign funds. And the world's biggest oil exporter has joined them: The Public Investment Fund has almost as many foreign assets as the Saudi Central Bank (SAMA).*

What's more, the available evidence suggests that the dollar share of these pools is in line with or higher than the dollar share of global FX reserves (it depends on the institution and jurisdiction).

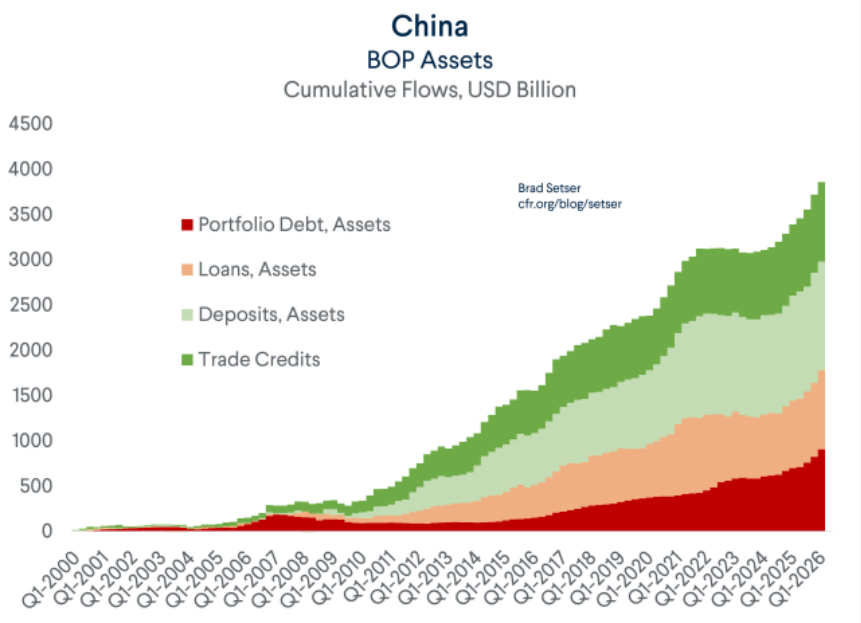


In other words, by looking at the (easily available) data on the world's static holdings of formal FX reserve assets, scholars and analysts miss most of the growth in the world's sovereign and quasi-sovereign foreign assets.

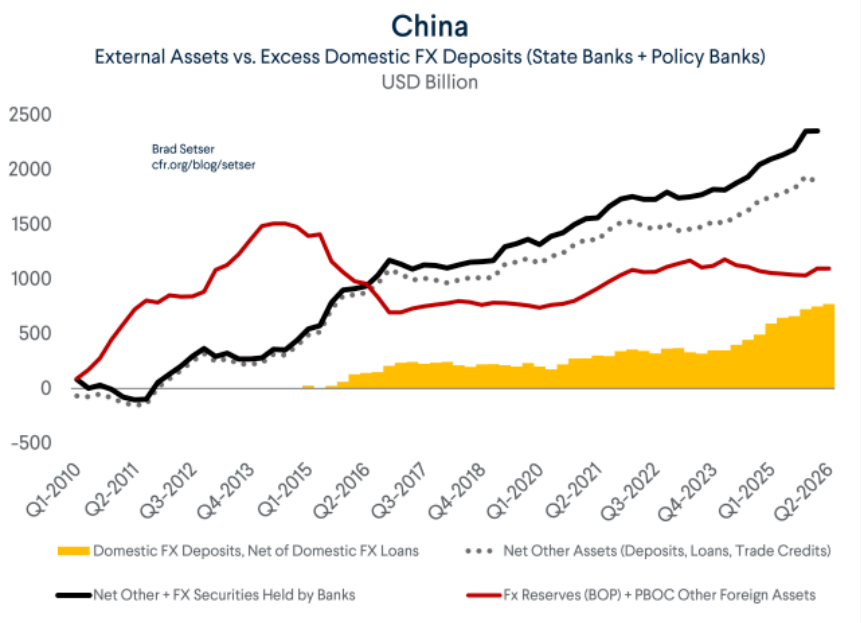
Consider China.

No serious analyst now disputes that China's state banks—including the policy banks (the China Development Bank, China Exim)—hold several trillion in foreign assets.

China reports $3.3 trillion in gross foreign assets to the BIS (the net position, counting foreign bank claims on the entire Chinese economy not just the banks, is $2.5 trillion). That maps to the BOP data, which shows almost $4 trillion in gross outflows through the banking system (technically, the sum of gross outflows in "other" plus the $500 billion in foreign currency bonds held by the state commercial banks)

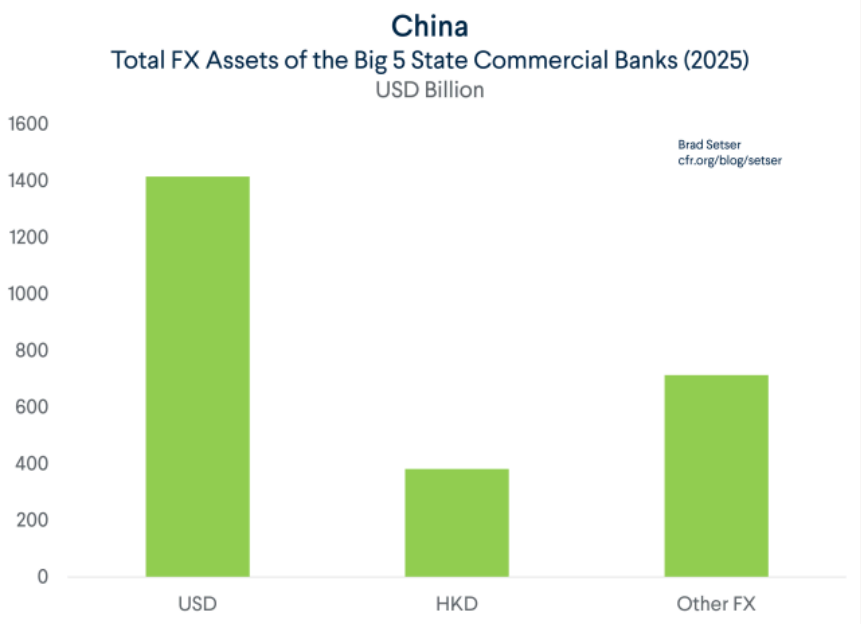


Since 2010 (and even more so after 2014) the net outflow through the state banks has exceeded reserve accumulation.



The broad contours of this story are confirmed by the balance sheet data reported by state commercial banks in their 2025 an ports, which showed

that the top five banks held a combined $2.5 trillion in foreign currency assets (mostly held abroad).



The Chinese haven't disclosed the foreign assets of the two policy banks (with at this stage the complicity of the IMF, which has neither analyzed the role of SAFE policy bank financing, nor highlighted the glaring gap in China's own reporting). But the work of AidData points to nearly $1 trillion in foreign assets, with a hefty dollar share.

The available data sources all suggest that the bulk of the foreign assets of the state banks are in dollars.

SAFE's disclosure—which covers the commercial banks—puts the dollar share of their offshore foreign currency assets at around 70 percent. That is well above the last disclosed dollar share of China's formal reserves (55 percent in). The banks' annual

reports suggest that about 70 percent of the banks' assets are in dollars and Hong Kong dollars (55 percent USD/15 percent HKD)—a bit below the dollar and Hong Kong dollar share of their liabilities (swaps out of the HKD fund a decent share of the banks euro assets).

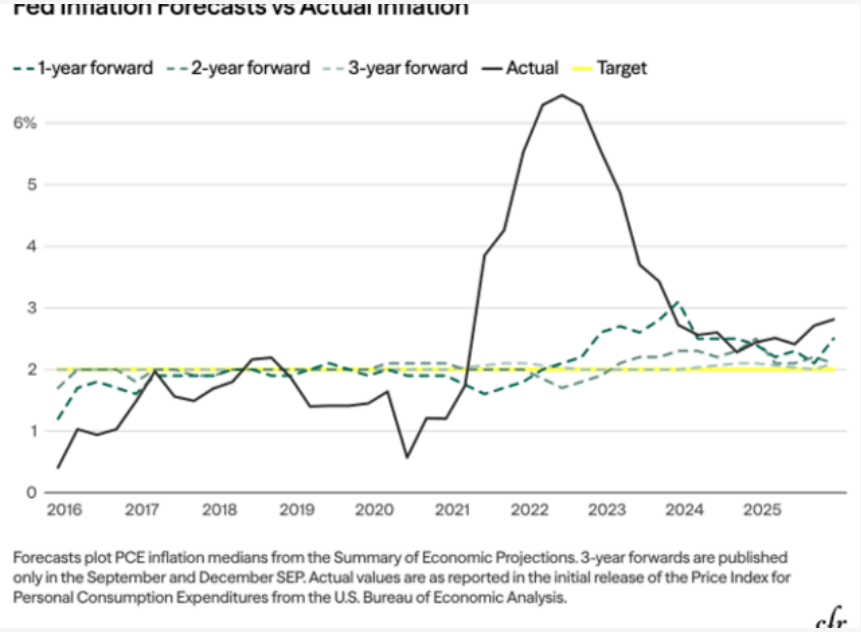
Put simply, SAFE's static dollar holdings aren't the important story.

The real story is the rapid growth of the state banks dollar holdings.

Japan holds a high (though undisclosed) dollar share in its FX reserves, and those reserves are primarily invested in Treasuries—so Japan's MoF is now clearly the largest contributor to the U.S. data on foreign official holdings of Treasuries. (SAFE has shifted its funds out of U.S. custodians, and thus increasingly appears in the data as a "private" holder in a European custodial center)

Japan's formal reserves haven't changed much in the last ten years; periodic sales to limit the yen's weakness have offset most of the MoF's accumulated interest income. Foreign currency reserves were $1.2 trillion in 2014 and $1.16 trillion at the end of Q1. They dipped to about $1 trillion at the end of August after sales of close to $100 billion at the end of July/start of August.

What has grown is the foreign portfolio of the GPIF. it was around $400 billion in 2014, ~$1 trillion in Q1 of 2026.

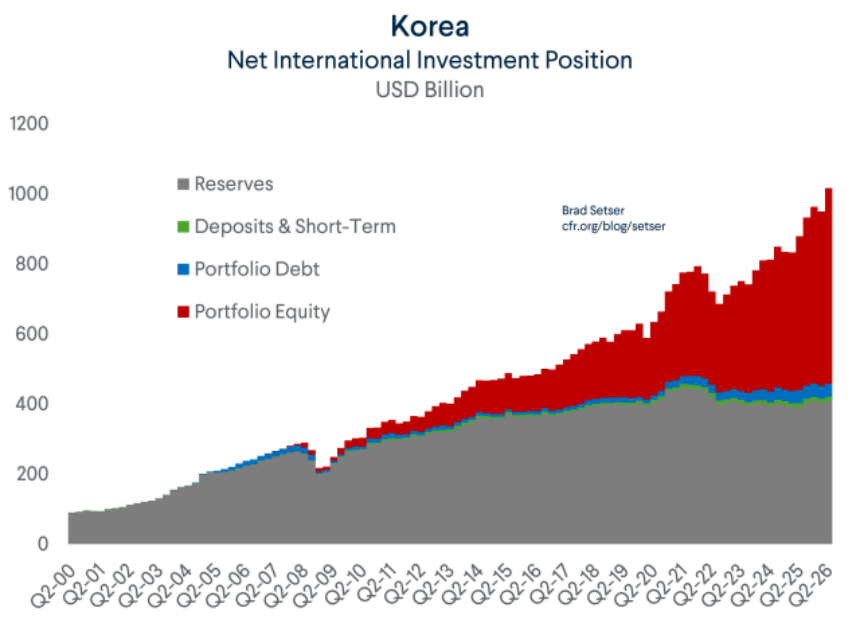


GPIF's foreign bonds aren't held exclusively in dollars, but roughly 52 percent of its foreign bonds (26 percent of all bonds) are USD-denominated.

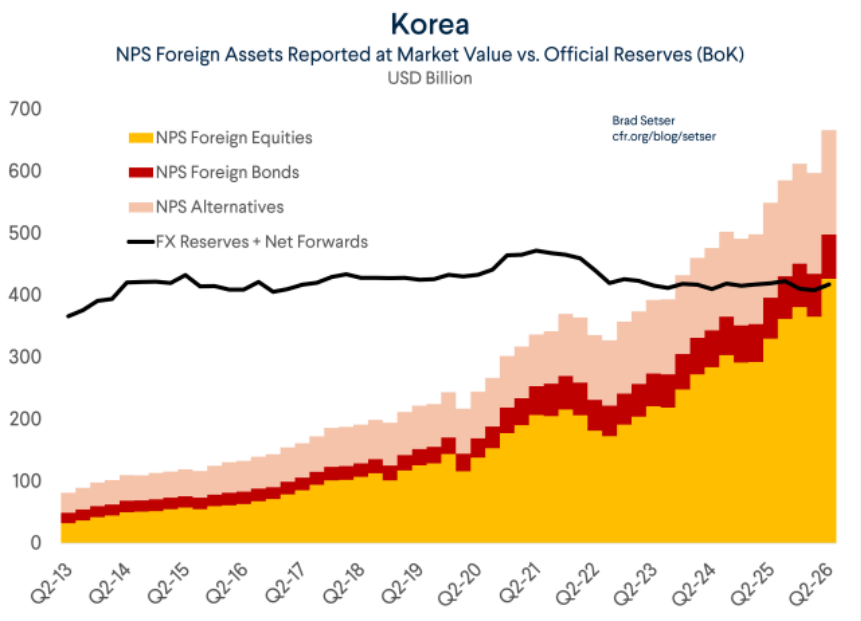
But the GPIF, like most international investors, holds a large share of its equity portfolio in the U.S. (~$165 billion, 65 percent of the foreign equity portfolio).

Korea is a more extreme version of Japan.

The Bank of Korea's FX reserves have been roughly $375-450 billion over the last ten years or so. Around two-thirds of those are now in dollars (as of the end of 2025).



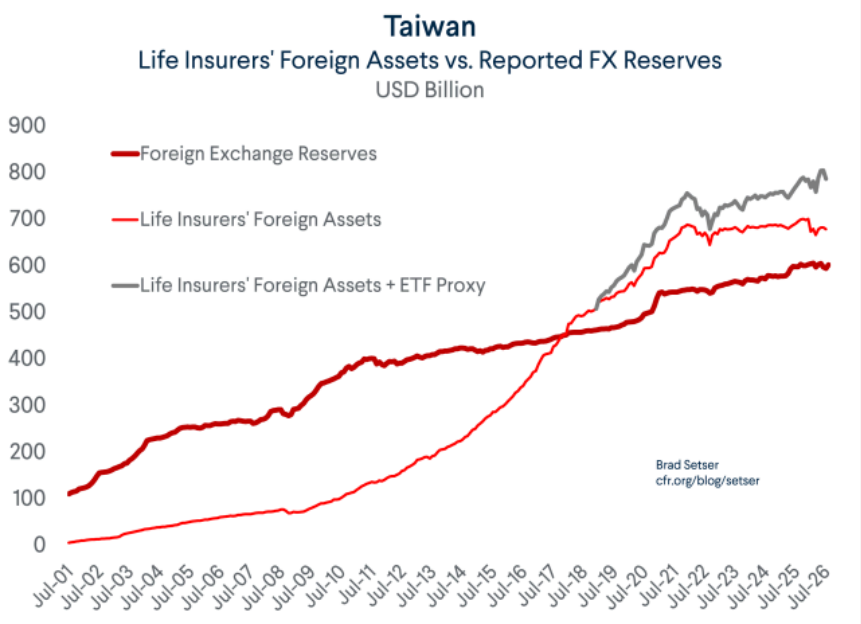
From 2012 to 2025, the foreign assets at the NPS have grown from ~$80 billion (around 21 percent of total assets) to over $660 billion (55 percent of total assets)—and something like 70 percent of those are in dollars. 76 percent of the fund's equities are in dollars, and 66 percent of its debt. And while a lower share of its "alternative" portfolio is in North America, most alternative mandates are awarded in dollars.



In other words, the majority of the dollar assets held by Korea's government (the NPS reports to the Ministry of Health and Welfare, and the Minister of Health chairs its asset allocation committee) aren't held by the central bank and aren't in the IMF's COFER data set.

Taiwan is an unusual case, as the life insurers aren't technically sovereign investors. But with their regulator now explicitly encouraging the them to hold an unhedged foreign currency book and hence they now rely on the state for their solvency, they have a bit of an official character (a significant appreciation of the TWD would blow through their volatility reserves and leave them with fewer assets than liabilities, as their foreign assets are largely held against TWD policies). The CBC certainly now uses the lifers desire to unwind existing hedges to help manage the Taiwan

dollar. And they have an exceptionally high dollar share in their foreign assets—over 95 percent.



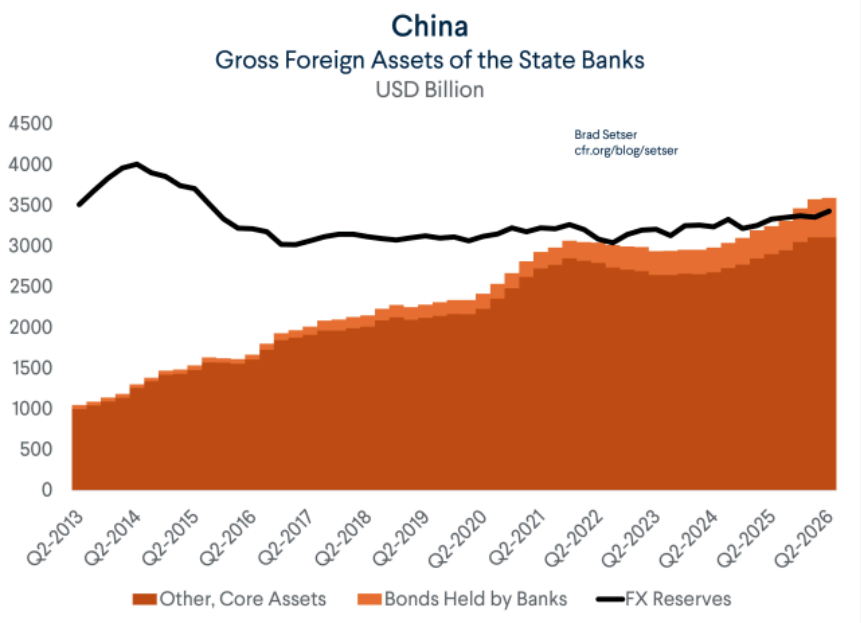
The Saudi Public Investment Fund (PIF) is another example.

It keeps roughly $200 billion of its $250 billion in foreign investments in dollars—well above the U.S. share of a standard global stock index. Almost all of the PIF's $30 billion in external debt is also in dollars.

And the PIF now has only slightly fewer foreign assets than the Saudi Central Bank has reserves.**

The broad story is thus pretty clear: the growth in the world's sovereign and quasi-sovereign assets is not coming through an increase in FX reserve holdings managed by the world's central banks.

And, in the critical case of China, the dollar share of China's quasi-sovereign foreign assets is likely higher than the dollar share of their formal FX reserves.



Adam Tooze has noted that inflows into the U.S. are now coming from investors looking for a profit, not investors looking for safety. What he calls the "profit dollar" manifests itself in a number of different ways. More inflows into corporate bonds and equities for one. Less official demand for Treasuries for another. More opacity too.

Those same trends are at play among sovereign and quasi-sovereign investors.

So don't obsess about the dollar's chare in formal FX reserves. Do recognize that the dollar's "reserve currency role" isn't the source of any significant new inflows into the dollar.

The dollar's dominant role in the international monetary system depends on much more than the size and composition of central bank FX reserves. It is as much, perhaps more so (given that FX reserve managers are themselves ultimately liability matchers) a function of the portfolio choice of a set of private/semi-private/quasi-sovereign investors that are much more difficult to observe. Reserve currency status is not only about FX reserves in a strict sense.

That's been true for some time. Most of the current flow into U.S. from the "official" sector is coming from investors who are not classic reserve managers.

Those flows remain heavily tilted toward the dollar, at least for now.***

And any real de-dollarization would likely occur first among these investors.

Put differently, the dollar's global role is increasingly as a source of returns, not a source of safety. The foreign bid, private and public, is for risk, not for Treasuries. That doesn't help Scott Bessent much right now, but it has helped keep valuations in the stock market extended. And the global debate on the dollar's role lags the evolution in the dollar's role, and the risks associated with that new role.

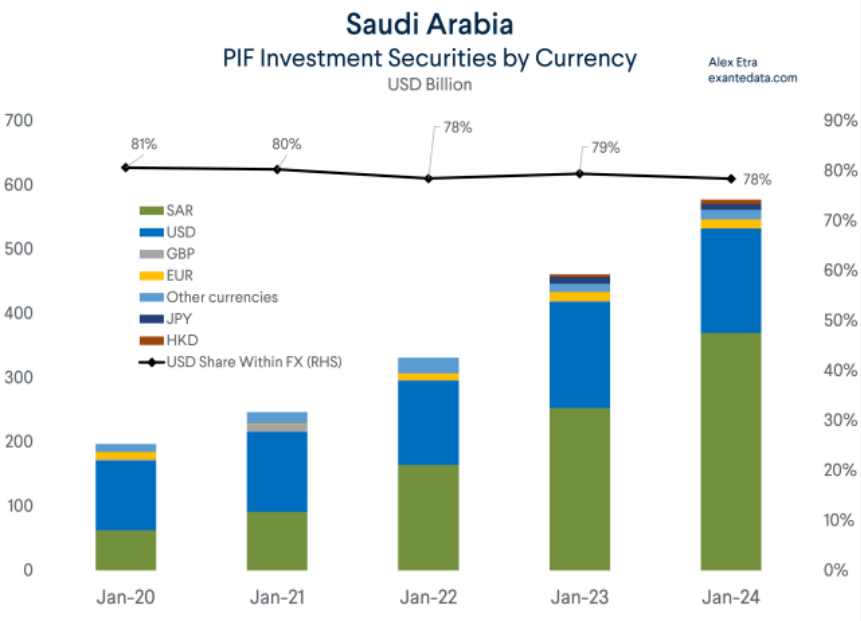
It is currently fashionable in some circles to point to the diminution of the "[illegible]nience yield" on

Treasuries even while the U.S. dollar continues to enjoy such a privilege. But the shift in the global investor base, and their apparent portfolio preference for U.S. risk assets in lieu of Treasuries, may help explain the divergence between the convenience yield on UST and the convenience yield on USD.

And of course, if the negative convenience yield implies that Treasuries are trading at a historical discount, the persistence of the latter—and its concentration in risk assets—suggests that the "profit dollar" itself is likely trading at a historical premium. That apparent premium on U.S. risk assets deserves as much (or more) scrutiny as the currency composition of central bank FX reserves.

* The PIF and Saudi Aramco also now have a certain amount of debt.

** Reserves (now mostly in Treasuries, so should appear in the TIC data, unlike at some times in the past.



*** Japan is the main exception; the GPIF's dollar share is below the likely dollar share of Japan's MoF.

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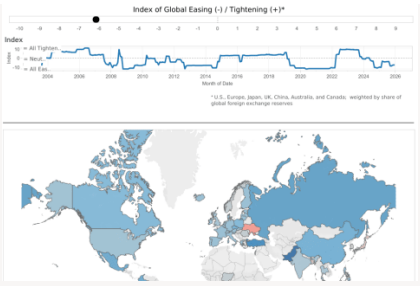
CAPITAL FLOWS BANKING



What's the Renminbi Got to Do With it?

By Brad W. Setser

October 11, 2026



Global Monetary Policy Tracker

By Benn Steil and Yuma Schuster

October 1, 2026



The U.S. Corporate Tax Base That Now Resides in Ireland

By Brad W. Setser and Michael Weilandt

September 3, 2026

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EXPERT TAKE FOLLOW THE MONEY

The U.S. Corporate Tax Base That Now Resides in Ireland

America's largest companies often pay more tax in Dublin than they do in D.C.



A statue of former Sen. Albert Gallatin stands at the Treasury Department in Washington, U.S., April 25, 2021. REUTERS/Al Drago

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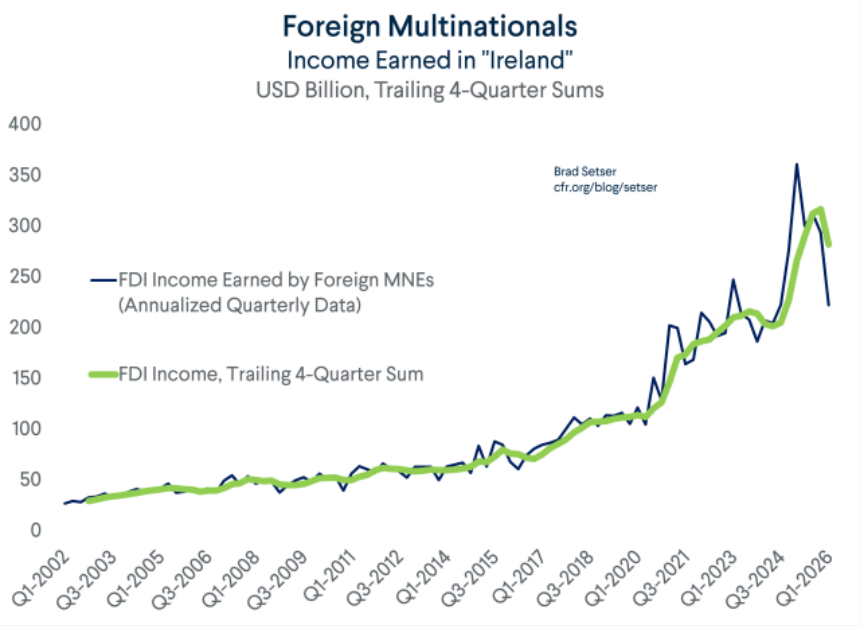


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This is a joint post with Michael Weilandt, a Research Associate at the Council on Foreign Relations.

Foreign multinationals, mostly American multinationals, booked just under $300 billion in profits in Ireland last year.



That isn't just because Ireland used to have a 12.5 percent headline corporate tax rate, and its current 15 percent rate on large firms is below the U.S. headline rate of 21 percent.

Ireland also permits deductions for qualifying expenditure on intangible assets, including the purchase of intra-group intellectual property, which can substantially reduce firms effective Irish tax rate.* And since Ireland is a member of the OECD and most firms now want to book their profits in "BEPS" compliant jurisdictions, Ireland is the center for corporate tax avoidance that isn't (quite) a pure tax haven.

The names of the firms that take the most advantage of lower tax rates in Ireland aren't exactly a secret.

Apple—widely thought to be Ireland's largest individual taxpayer thanks to its mostly Irish $80 billion "global" profit—hasn't reported the amount of cash it paid the Irish government in 2025.

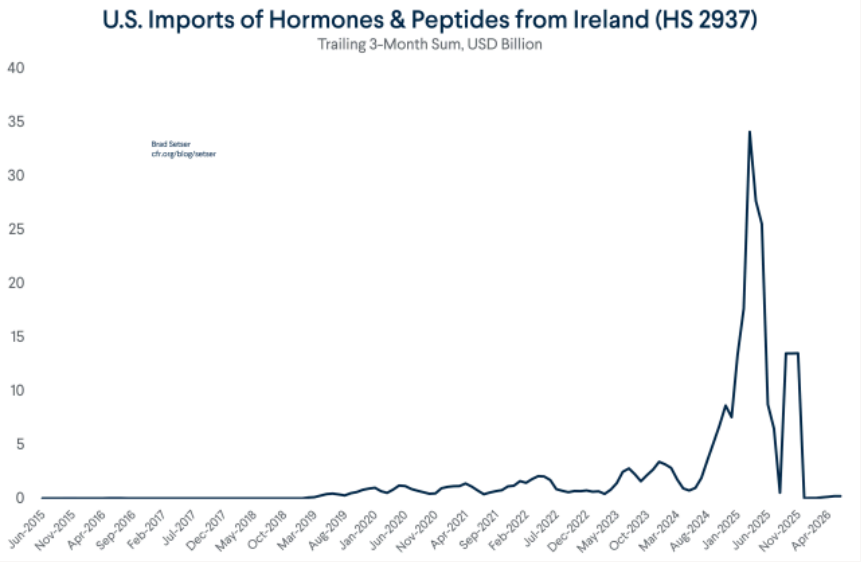
Microsoft has reported the tax it pays the Irish government. Per its latest 10-K (p.75), in FY 2026, it booked over 80 percent of its $62.5 billion international profit in Ireland. Microsoft also reports that it actually paid $6.5 billion to the Irish government (and another $5.5 billion in tax in other foreign jurisdictions), more than the $6.25 billion it paid to the U.S. Treasury. Paid here means a transfer to the Irish treasury, not a set aside to cover expected FY 2026 taxes.** It set aside $12 billion to cover expected 2026 foreign tax payments, and $2.5 billion for expected U.S. federal income tax.

Eli Lilly, like Microsoft, paid more in tax to Dublin than to D.C. Eli Lilly's 10-K states that it paid $6.6 billion in tax to Ireland in 2025, and only $3.3 billion to the U.S.

Now, Eli Lilly likely also played a bit of a tax game in (calendar year) 2025, as its Irish tax payments were unusually high.

Lilly and the other big pharmaceutical companies shipped to the United States back in the first quarter of 2025. Back then the pharmaceutical companies were petrified that President Trump might actually impose real tariffs on profitable American firms playing tax games abroad. Those tariffs though never materialized. Firms did "deals" with the Trump administration instead.

But the threat of tariffs had a huge impact on the trade data—Lilly rushed to produce Mounjaro's active ingredient in its Irish factories, and it then rushed the product to the U.S. ahead of the tariff deadline (there is a six-digit line item in the trade data that is more or less Mounjaro, for the curious). And it seems like Lilly's Irish subsidiary booked the profit on its sales to Lilly U.S. ahead of any actual sale to U.S. consumers.***



The net result, obviously, was a big Irish tax liability. And a big windfall for Ireland's Treasury, as Lilly displaced Pfizer as Ireland's third largest corporate tax revenues.

The $15 billion a year that these firms generally pay in tax in Ireland is real money, as they say (they paid €13 billion in 2024 according the Irish fiscal council; the 2025 number will be out soon).

Ask the top brass at the Pentagon. $15 billion a year is probably three new attack subs, or five Arleigh Burke class destroyers (think mobile air defense systems). Or for several batteries of Patriots (if Raytheon can make them). Had those funds been paid to the U.S., it would not quite cover the cost of an aircraft carrier (costs have increased recently) but it would have come close. $30 billion over two years is a carrier task force. Enough, in other words, to make the tradeoff between g[redacted]l butter a bit less acute.

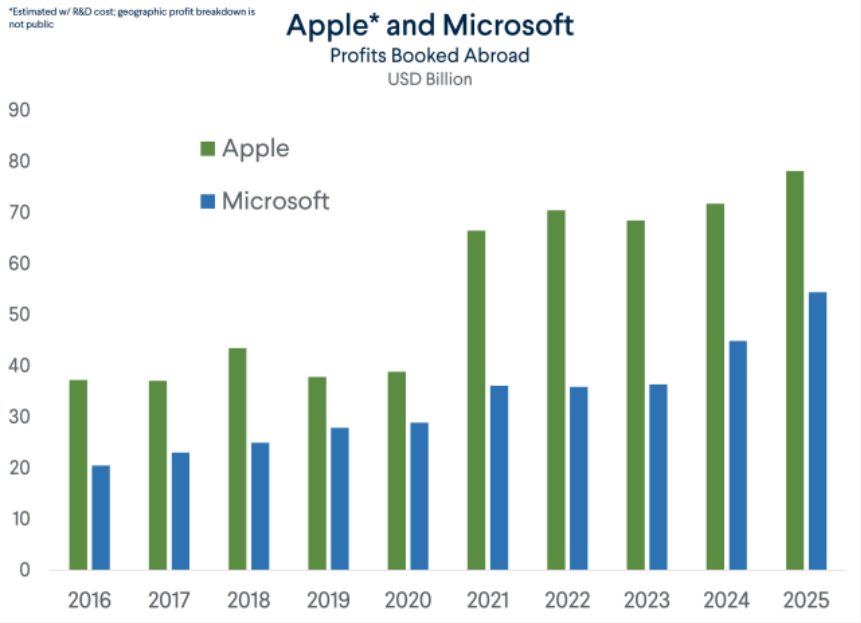
That is simply from tax that firms are paying in Ireland on the profit from intellectual property largely developed in the United States.

And profit shifting remains a real problem.

That is what the Irish data from 2025 shows us. And there is no sign that 2026 will be much different.

A lot of that is Apple and Microsoft.

Both have global profits will in excess of the total profits of all the large American pharmaceutical companies.



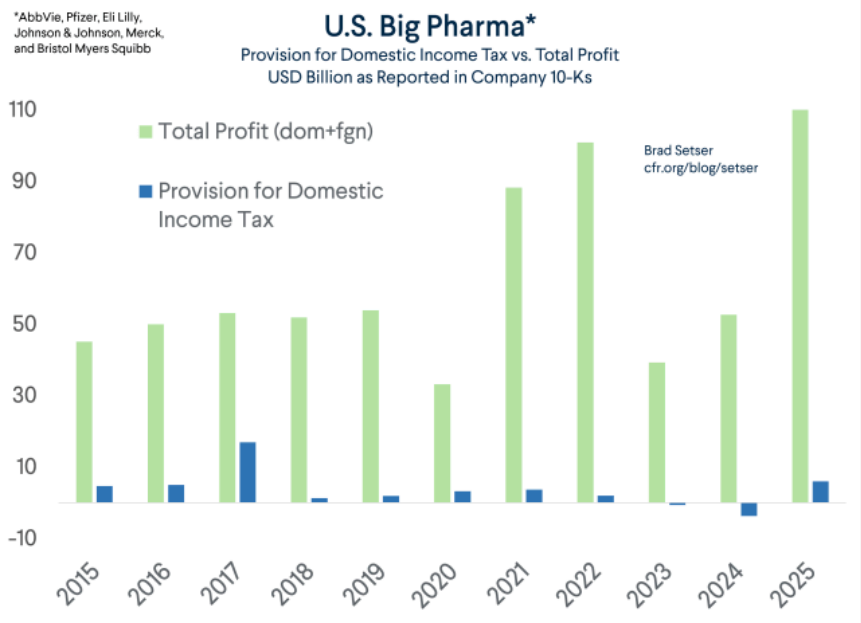
But Apple and Microsoft are essentially moving the profit from their foreign sales to Ireland. They still pay a decent amount of U.S. tax (as do Alphabet and Nvidia, which both U.S[illegible]ed their intellectual

The tax game of "Big Pharma" (the top six U.S. pharmaceutical companies by revenue) is a bit different. Most aggressively try to move the profit on their U.S. sales out of the U.S., which generally has required offshoring both their intellectual property and the actual production of the active ingredients for their most profitable drugs.

Back in 2023 and 2024, the top six U.S. pharmaceutical companies didn't set aside anything to pay expected U.S. tax. They have pointed out (to me, and to Senator Wyden) that they did make cash payments to the U.S. Treasury, but those payments were the settlement of the legacy tax liability on their pre-2017 profit (don't ask, it is complicated—the key words are deferral and deemed repatriation).

In 2025, the major pharmaceutical companies did set aside a few billion dollars to pay U.S. tax. Eli Lilly doesn't seem to be quite as aggressive in sweeping profits out of the U.S. as its peers; it has consistently reported setting aside a bit of money to pay expected U.S. taxes on its global profit. Johnson and Johnson, which historically has swept all its profit out of the United States, also had to set aside funds to pay U.S. tax as part of some litigation related tax gymnastics.

But make no mistake, Big Pharma continues to engage in aggressive forms of tax avoidance —and still pays almost nothing in U.S. tax. See the chart below



And there is no evidence—at least, none that we see—that suggests any of the Trump deals have changed the core tax structure of big pharma.*****

The top six generally still report losing money on their U.S. operations.

The big companies also generally report that the U.S. accounts for most of their revenue.

And they generally report that they make most of their money offshore (particularly if "special situations" like corporate reorganizations, mergers, and litigation set asides are left out). Just look at the firms' own 10-Ks.

AbbVie reports losing $3.5 billion in the U.S. and making $10 billion abroad.

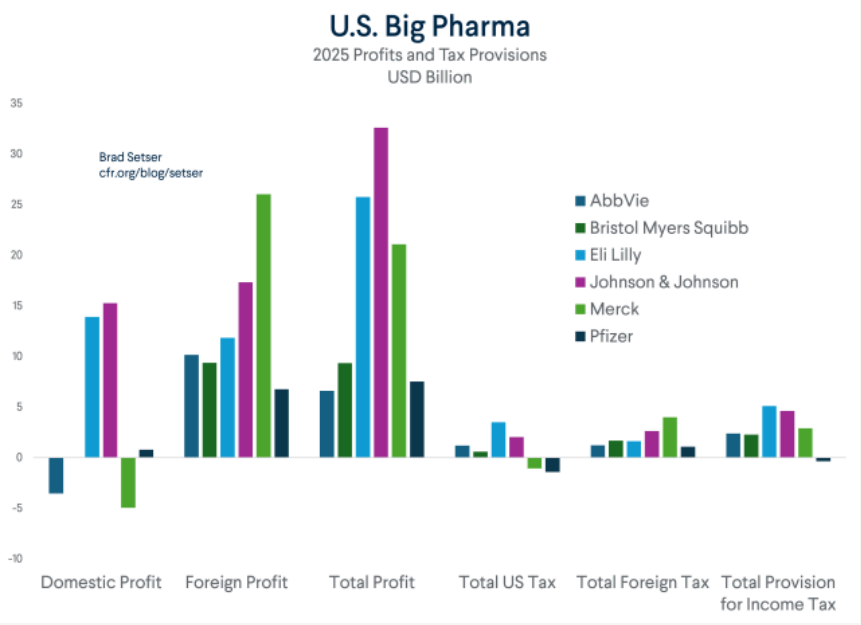
Pfizer reports just $776 million in U.S. profit and almost $7 billion abroad.

Bristol Myers Squib reported a $19 million loss in the U.S. and $9.3 billion profit abroad.

Merck reported a $5 billion loss in the U.S. and $26 billion profit abroad.

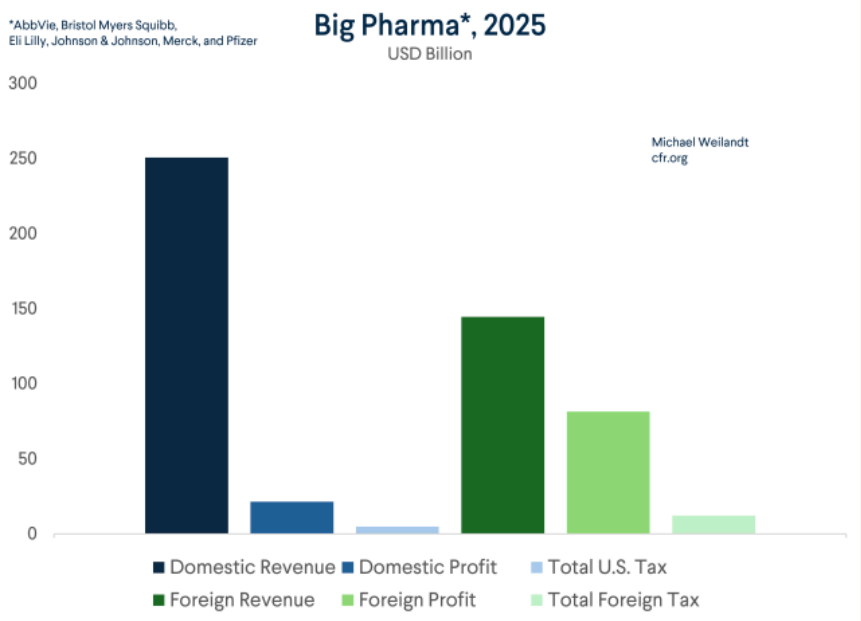
Johnson and Johnson would be the same in a typical year. Back in 2024, it reported a $500 million loss in the U.S. and $17 billion profit abroad, but its 2025 tax liability was driven by the reversal of some litigation set asides (see page 69 of J&J's latest 10-K).

Lilly seems to have a more balanced tax structure: in 2025, it reported making $14 billion in the U.S. and $12 billion abroad.

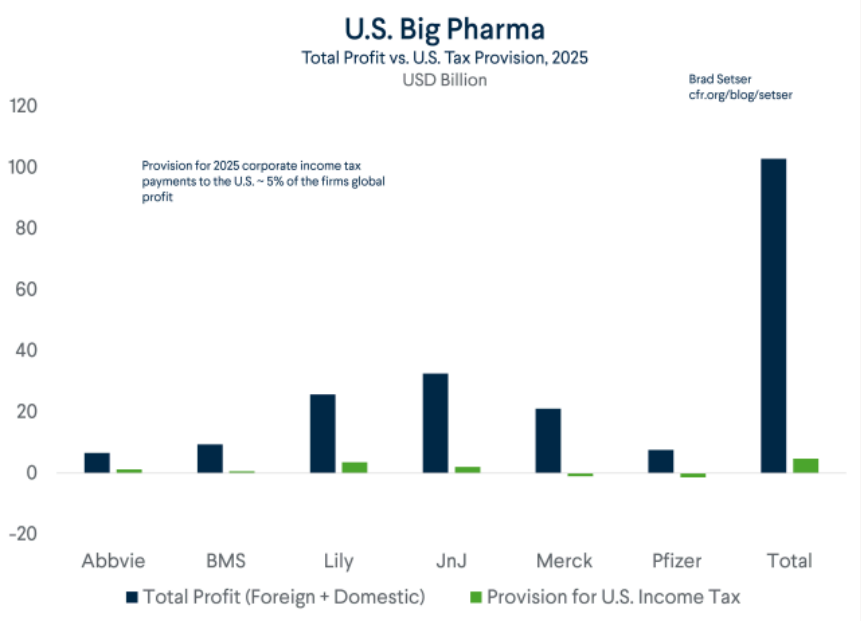


As a result, the leading U.S. pharmaceutical companies pay more in tax abroad than in the United States, with several firms setting aside nothing to cover their expected U.S. income tax liability.

In a normal year, without any corporate reorganizations or litigation related profits, only one of the top six would set aside a meaningful sum to pay U.S. tax. So, in a typical year, there would something like $100 billion in profits and only a few billion set aside to pay expected U.S. tax.

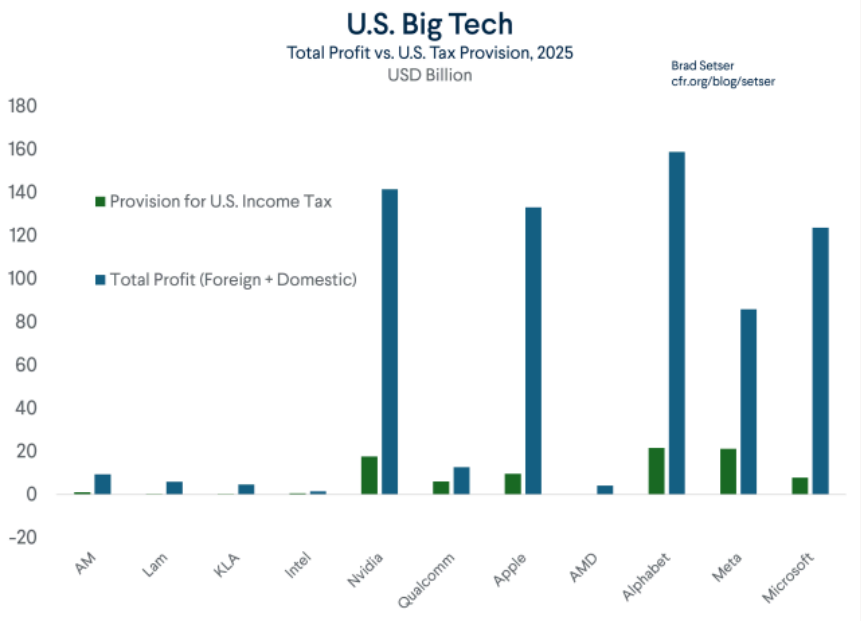


This shows up in a couple of easily verifiable plots.



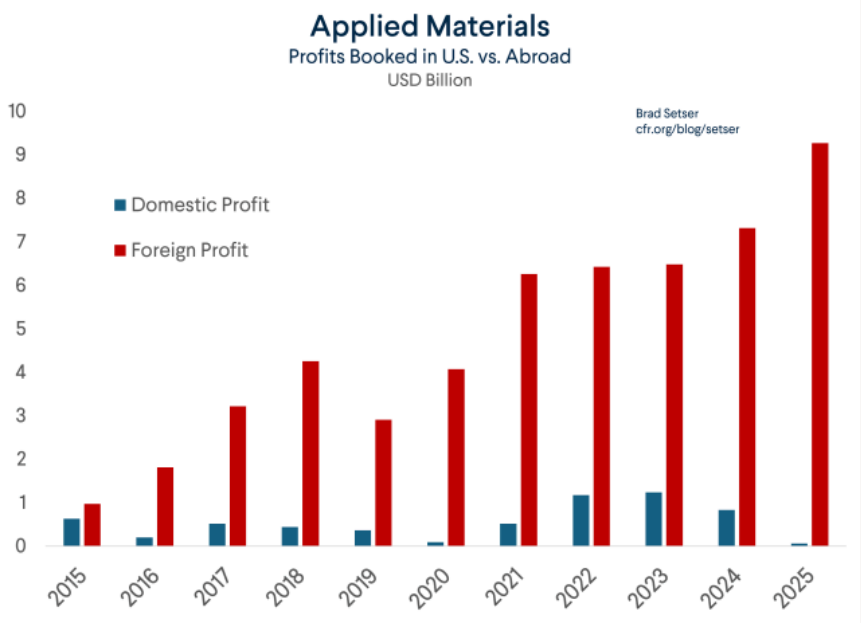
While Big Pharma is the most obvious case, the incentive to offshore jobs, production, and profits isn't limited to one sector.

In the technology sector, for example, a number of firms set aside much less to cover expected U.S. corporate tax payments than would be expected given their large global profits.

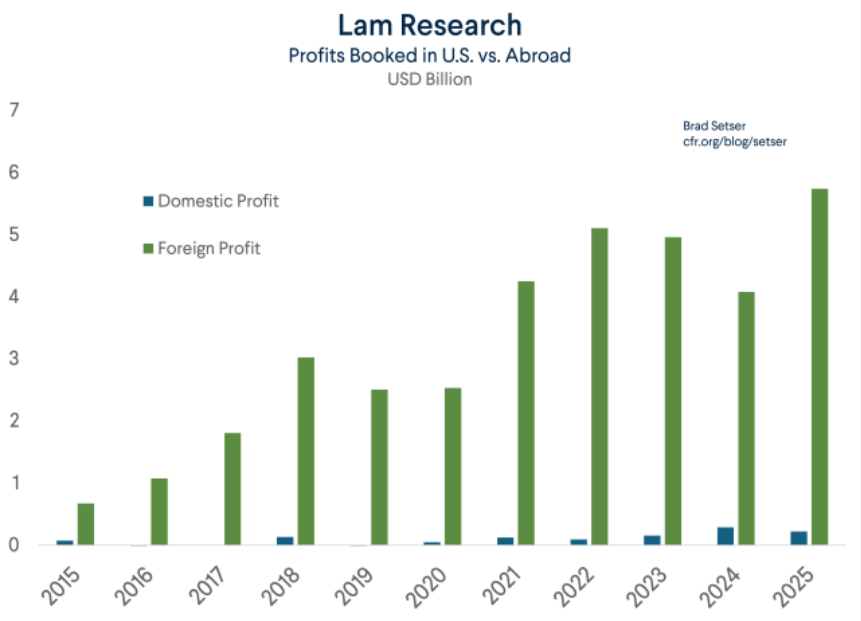


Take a very strategic sector, the production of the equipment used to manufacture semiconductors.

Applied Materials reports making all its money abroad (almost certainly in Singapore, as noted in its 10-K, where Applied Materials "have been granted additional conditional reduced tax rates" which expire in fiscal 2030) and almost nothing in the U.S.



Lam Research also reports making all its income abroad, using Malaysia as its tax hub (from its 10-K: "Effective from fiscal year 2022, the Company has a 15-year tax incentive ruling in Malaysia for one of its foreign subsidiaries.")



KLA is a bit more patriotic; it actually pays a bit of U.S. tax.

To be sure, none of this is in any way illegal. Every firm can honestly report that it pays all the tax that it is legally due.

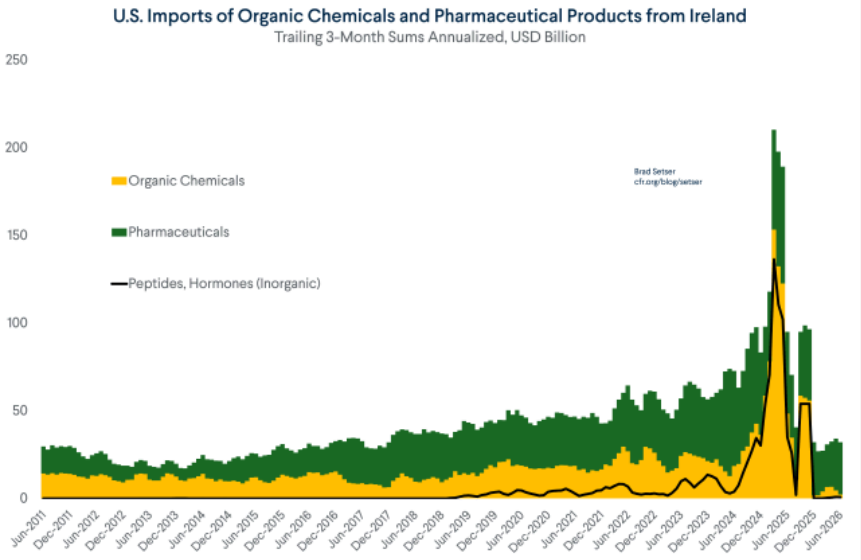
The issue is the law, which continues to create "pay tax in America last" incentives, and often "produce in America last" incentives too.

This can and should be changed, likely through a bipartisan coalition that unites around a patriotic tax agenda.

* Most U.S. big pharma firms are still paying a bit of cash to the U.S. Treasury to settle their deferred international tax liability from the ere Tax Cuts and (Irish) Jobs Act; the cash tax paid thus exceeds the amount they set aside out of current profit for expected current tax. A few firms have tried to use this fact to obscure the fact that they don't think they owe anything to the U.S. federal government out of their 2023 or 2024 profits.

** Alphabet and Nvidia benefit from the lower tax rate on foreign profits in the U.S. tax code; they aren't paying tax on their foreign profits at 21 percent. But because their IP is in the U.S., the U.S. government gets the lion's share of the tax paid. When the IP is offshore, the U.S. only gets a small top up tax.

***



**** Large groups with annual consolidated revenues of at least €750 million are subject to a 15 percent minimum effective tax rate under Pillar Two.

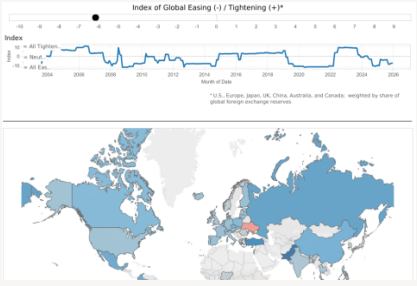
***** Gilead is much less aggressive in its profit shifting.

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IRELAND PHARMACEUTICALS AND VACCINES



Global Monetary Policy Tracker

By Benn Steil and Yuma Schuster

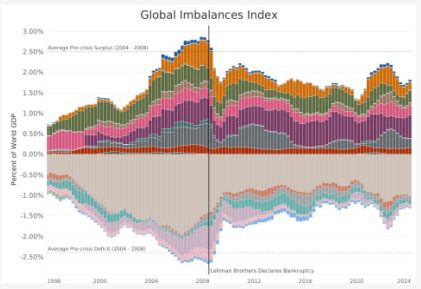
October 1, 2026



De-reservification, Not De-dollarization

By Brad W. Setser

September 15, 2026



Global Imbalances Tracker

By Benn Steil and Yuma Schuster

August 12, 2026

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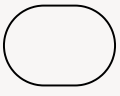
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EXPERT TAKE FOLLOW THE MONEY

The Renminbi Appears to be Undervalued by 30 Percent

China's official current account data understate the scale of its external surplus and, by extension, the degree of renminbi undervaluation.



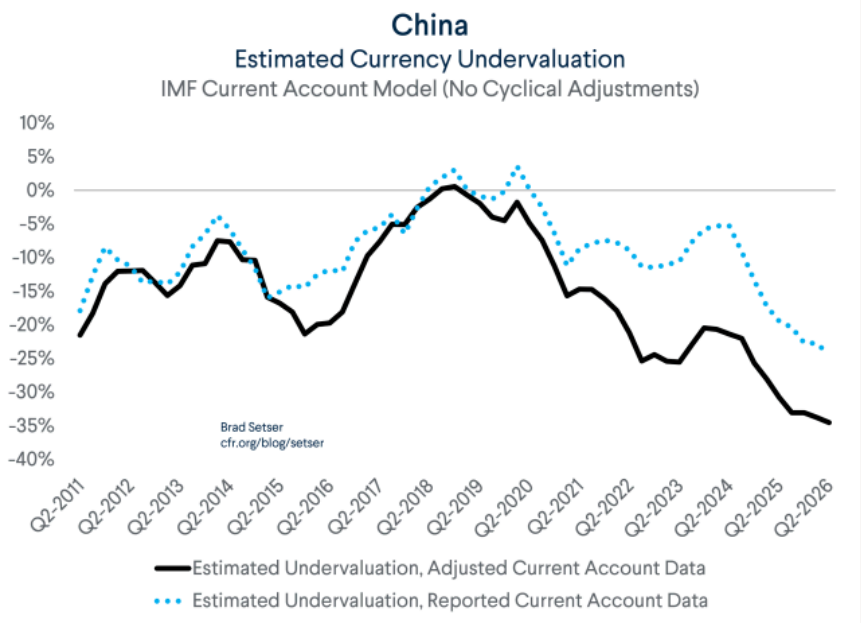
PUBLISHED

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Brad W. Setser CFR EXPERT
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Harvard's Dr. Gopinath (formerly of the IMF) asked about the basis for my estimate (in Greg Ip's excellent WSJ column) that the RMB is now undervalued by 30 percent.*



The simple answer: I used the IMF's own current account methodology but didn't take China's reported current account surplus at face value. I believe that the surplus is significantly understated, as it doesn't map to other known numbers about China's economy and trade performance.**

The IMF's current account gap methodology hinges on the difference between the observed current account and the "current account" norm calculated by the IMF (on the basis of demographics, levels of development, the net international investment position, and other variables). For China, that current account norm is now estimated at 0.6 percent of GDP (2026 ESR, Table 2.6)

A cyclical adjustment added 0.2 percent of GDP to the estimated norm for 2025. The end-2025 surplus was 3.8 percent of GDP. The 3 percent of GDP gap resulted in an estimated undervaluation of 21 percent as the IMF's work implies that a 7 percent move in the exchange rate (or equivalent changes in other policies) reduces the surplus by a percentage point of GDP.

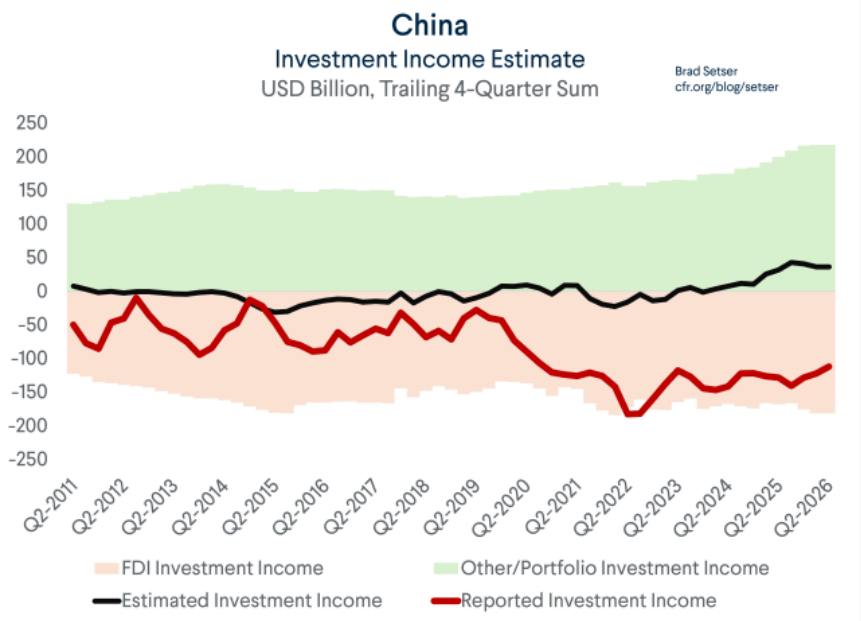
A bigger surplus thus implies a bigger "gap" and a bigger estimated undervaluation (for a given constellation of policies).

And I think the 3.8 percent of GDP surplus for end-2025, now up to 4 percent of GDP in the last four quarters of data, is understated. I am not the only one who thinks by the way; the raw customs data—adjusted for the observed deficit in travel services and the recent surge in gold imports—implies an overall surplus of close to 6 percent of GDP these days.

There are three reasons to believe that the reported current account surplus understates the scale of China's real surplus:

1. ***The reported investment income deficit makes no economic sense.***

The reported deficit on investment income is very hard to understand. You don't have to take my word on this. The IMF looked into China's investment income balance and agreed (2025 ESR, Box 1.1). So did Ma and Wei.



The fact that China reports an investment income deficit of over $100 billion with net foreign assets of over $4 trillion isn't easily explained. Yes, there can be a difference in return on foreign direct investment in China and Chinese direct investment abroad. But with roughly equal leve vestment in both

directions, that gap in returns on direct investment would need to be between 6 and 8 percentage points to offset the $120-160 billion in interest income that would be expected on China's net holdings of foreign bonds and other interest-paying assets at a conservative 3 to 4 percent interest rate (CDB and state commercial bank loans, if anything, should carry a higher rate, as the CDB lends at a big markup over the U.S. policy rate).

And there is no real evidence that foreign direct investors are earning a ton of money inside China these days. The auto companies certainly are not. Many have abandoned their Chinese JVs, or are using them primarily to supply global markets. The exception—Apple—actually proves the rule. Apple does have a nice business in China. But China's 2022 statistical adjustments should move Apple's returns out of the "direct investment" line as Apple's IP returns accrue to its Irish subsidiary, not its Chinese subsidiary, and into the balance of payments as a small goods deficit.***

The evolution of the investment income line item also doesn't make any sense. The investment income deficit increased a lot in 2019 and 2020. Certainly, interest rates were low globally in 2020. But that doesn't explain why the investment income balance remained in a large deficit even after global interest rates soared. The returns on China's large and growing

external loan and bond portfolio should have soared, but they didn't. That makes no sense.

China in Q1 reported something like a 2 percent return on its foreign bonds and overseas lending. That doesn't compute, as SAFE eked out a 3 percent return in the low-for-long era after the GFC and can get more than that from a portfolio of five-year Treasury notes and five-year bunds.****

2. ***China started importing a ton of gold in 2026.***

China's gold imports have gone from around 0.5 percent of GDP to 1.5 percent of GDP in a couple of quarters. That is strange—especially as China's rising interest in gold came as gold prices fell.

Gold is a strange import. If the gold isn't going directly to the central bank, it counts as an import even though it is often held as a financial asset. And if the gold is going directly to the central bank, it is counted as the purchase of a financial asset. Some even think SAFE is buying gold through domestic brokers as a means of shrinking the reported surplus. It thus isn't really controversial to move the gold import line out of the current account and over to the financial account.

3. ***China started fudging its balance of payments goods number when it changed its balance of payments methodology in 2022***

That change is now statistically faded into the 2021 data as well.



The changes were in theory an attempt to implement the IMF's latest guidance (BPM6, which the U.S. hasn't done). While the IMF has recognized problems with China's reported investment income deficit, the IMF statistics department hasn't given an inch on the new goods data, and argues that China, unlike the U.S., has at least tried to follow its statistical guidance.

But China's actual implementation of the IMF's guidance was suspect to say the least. I covered this debate in detail a year ago (Apple plays a large role in China's argument). And the IMF refuses to assess whether the results of the new methodology are really an improvement over the results of the old methodology.

One of the adjustments that SAFE outlined to the IMF back in 2024—the "in China for China" adjustment for firms that don't have an onshore JV and rely on contract manufacturing—is in principle reasonable. Moving Apple's profit out of income over to the BOP goods line would be consistent with Apple's treatment in the Irish balance of payments data, but it should have produced a $20-$30 billion change in the goods number, not a $300 billion change. And moving Apple's "in China" profit from one line of the current account to another shouldn't have changed the overall current account numbers.

The other adjustment, which arbitrarily reduces exports in the balance of payments down relative to customs on a never demonstrated claim that contract manufacturers were inflating their exports in the customs data by using a firm's global wholesale price in customs rather than the amount actually received, simply doesn't check out. The IMF, to the best of my knowledge, has never looked into the details here either.

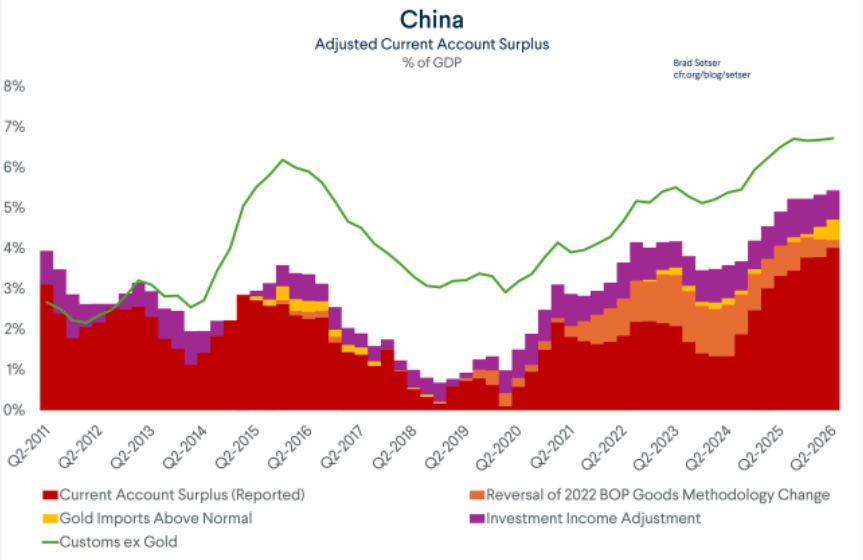
It is relatively easy to construct a consistent time series using China's pre-2021/22 methodology for calculating its balance of payments goods data (the standard adjustment moves insurance and freight charges over to services). The new methodology, at least until recently, really did lower the goods surplus, and thus the overall current account surplus.

There was clearly a massive downward adjustment to balance of payments goods for a while, without offsetting changes in other lines in the current account.

I was ginned up about this for a while. It really had a big impact on the 2023 and 2024 current account surplus numbers and thus changed the IMF's overall assessment of China. But—for complex reasons—the impact of this adjustment has shrunk, and it is now less significant than the investment income and gold distortions.***** See the grey and gold bars in the chart below.



When these factors are taken into account, I estimate that China's surplus should be around 5.5 percent of GDP—which, without the cyclical adjustment, works out to a 35 percent undervaluation (33 percent, with the cyclical adjustment).



I also used the IMF's methodology to estimate what the undervaluation should have been in the past (without the cyclical adjustment, as I cannot estimate that reliably).

The larger estimated undervaluation checks out against a range of other indicators.

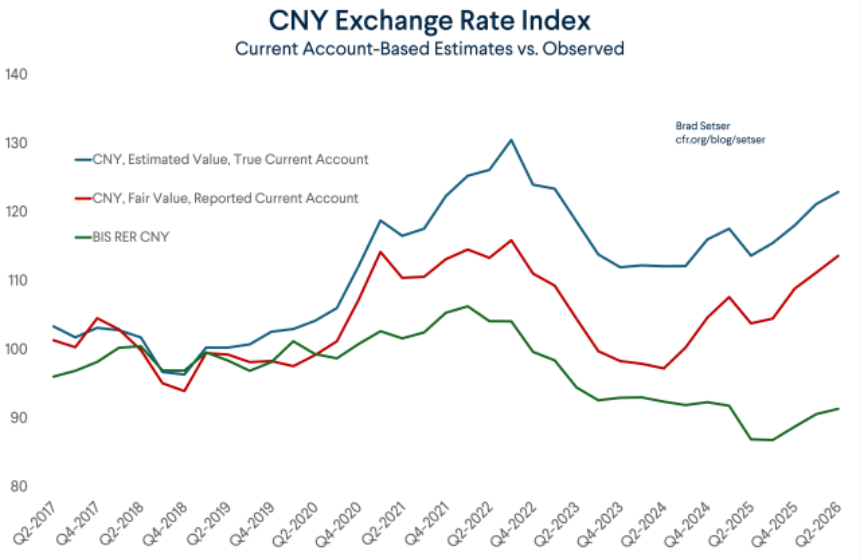
The Big Mac Index (37 percent undervaluation).

Industry-specific cost comparisons. Greg Ip has a nice chart from Goldman. European auto analysts think China has a 30 percent edge in production costs.

It is also clear in China's export outperformance relative to global trade. And the breadth of this outperformance.

At the end of the day th[illegible]h, it doesn't currently

undervalued by 20 percent or 35 percent. There is consensus that there is a significant undervaluation.



Fixing that is, by far, the easiest way to address concerns about China's ever-rising trade surplus—and to get Xi to reconsider China's growth model and do more to support internal demand.

* Dr. Gopinath observes that the IMF's current account gap can be closed through a combination of exchange rate moves, domestic policy changes and changes in the international environment (i.e. slower demand growth abroad, slower growth in global trade), not just through exchange rate changes. That is a fair point. Though I would note that there isn't much evidence over the last 15 years that Chinese demand growth spills over into import demand.

formally, income elasticity that is *way* below one. That implies the exchange rate, which does seem to impact export performance more than the IMF believed in the past, will need to do much of the work.

** The customs surplus can (and has been) checked against counterparty data. The travel deficit now broadly maps to observed tourism flows. And the reported NIIP is smaller than it should be given China's accumulated surpluses. The gross asset and liability position in the IIP data can nonetheless be used to generate an independent estimate of investment income (part of the current account).

*** I agree, for what it is worth. My concern is that China never showed its work, and there isn't any evidence that moving Apple's "in China" return over to the BOP goods line items was matched by offsetting adjustments to either the services line (if the "in China" profit was previously counted as a royalty to Apple Ireland) or the income line (if it was previously added to the income line, adding to China's deficit in FDI income). The current account impact of moving around Apple's $20 to 30 billion "in China" profit should be zero.

**** The duration on China's visible holdings of Treasuries is a bit above the norm for reserve managers, and China holds a ton of U.S. Agency bonds (MBS), so it should get an above average

return on its dollar reserves. SAFE isn't particularly risk averse.

***** I would like to think my efforts here had an impact on SAFE, if by nothing else making folks aware that there was an issue. But in reality, SAFE appears to be adjusting the BOP goods line to limit reported errors, and as the renminbi stabilized, settlement conversion increased, and the state banks started adding more to their financial assets, SAFE needed to report a higher current account surplus to map to the larger observed buildup of Chinese assets abroad.

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